

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Wi-Fi HaLow chip with Edge AI

Methods2Business has introduced a Wi-Fi certified HaLow chip with integrated Edge AI. The chip combines Wi-Fi HaLow and Edge AI into one solution. It operates in the sub-1GHz band, enabling long-range wireless communication with low power



use. By combining long-range connectivity with

on-device AI processing, the chip allows real-time data analysis. Devices can detect issues, predict failures, and take immediate action without sending raw data to the cloud. The sub-1GHz frequency enables signal penetration through walls and obstacles while reducing power consumption. This design supports IoT systems for various industries, including factories, agriculture, infrastructure, logistics, and healthcare. It lowers network latency, cuts data transmission costs, reduces reliance on internet access, and enhances data privacy and responsiveness.

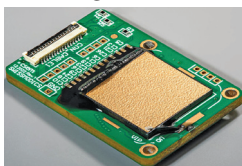
Methods2Business

<https://www.methods2business.com>

Image sensor that converts X-rays into electronic signals

CSEM and QDI Systems have developed an image sensor that converts X-rays into electronic signals using quantum dots on a CMOS platform. A Spray-coated layer of quantum dots, tuned to absorb specific wavelengths, is applied to CMOS chips. This design removes the need for components like scintillators, hybrid bonding, or

materials such as InGaAs. It reduces size and cost while maintaining sensitivity across X-rays, visible light, and short-wave infrared (SWIR). High wavelength sensitivity and fill factor



preserve pixel density and power efficiency. The technology enables imaging in fields where it was previously costly or impractical. In healthcare, it supports X-ray and mammography systems with reduced radiation exposure. In industrial inspection, it supports the testing of aircraft and machinery. In agriculture and recycling, it enables SWIR imaging for crop monitoring and material sorting.

CSEM and QDI Systems

<https://www.csem.ch/en/> and

<https://qdisystems.com/>

Compact, low-power scanning kit

OQmented has introduced the HYPE-RION evaluation kit. The kit uses the HYPERION BubbleMEMS mirror with a 1.3mm diameter and an optical field of view from 15° to 70° diagonal. The mirror operates with multi-interlaced



laser scanning at 35kHz and 19kHz. The kit supports automatic power-up and operates on a 3V to 5V supply. The

design reduces size and weight while maintaining scanning precision and power efficiency. Vacuum packaging protects the system and supports long-term stability. Developers can build smaller, lighter products without losing scanning performance. Wearable and

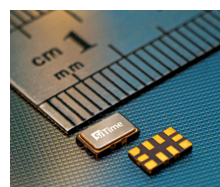
portable devices gain battery life due to low power use.

OQmented

<https://oqmented.com>

Single-chip timer for AI compute nodes

SiTime Corporation has introduced the SiT5977 Super-TCXO, a single-chip timing solution for AI compute nodes that need bandwidth and network



synchronisation. Supporting Smart NICs, acceleration cards, switches, and compute nodes, the

SiT5977 delivers 3x better synchronisation and supports 800G network speeds while using 4x less space than other designs. By replacing multiple timing components, it simplifies system architecture, allowing designers to fit larger processors or add more functions. This synchronisation improves data transmission, reduces AI accelerator idle time, and allows full use of 800G bandwidth, leading to higher performance, lower power use, and reduced cost for data centres.

SiTime Corporation

<https://www.sitime.com>

Local-processing neuromorphic microcontroller

Innatera has launched Pulsar, a neuromorphic microcontroller that combines spiking neural networks with signal processing, integrating a RISC-V CPU, CNN accelerators, and FFT engines to process sensor data at the source. Local processing removes external compute and cloud dependence, cutting energy